Jenny T. Liang

jtliang@cs.cmu.edu • jennyliang.me • liang-jenny • jennytliang

Education

Carnegie Mellon University, Pittsburgh, PA, USA — 2022 –
Ph.D. in Software Engineering
Advisor: Brad A. Myers

Carnegie Mellon University, Pittsburgh, PA, USA — 2022 – 2024
M.S. in Software Engineering

University of Washington, Seattle, WA, USA — 2015 – 2021
B.S. in Computer Science, B.S. in Informatics

Publications (* denotes equal contribution, Google Scholar)

Conference Publications

C15 **Mapping the Design Space of User Experience for UI Control Agents**
R. Cheng, J.T. Liang, E. Schoop, J. Nichols
ACM Conference on Intelligent User Interfaces (IUI) 2026

Oral Presentation (top 5% of accepted papers)
C14 **EDIT-Bench: Evaluating LLM Abilities to Perform Real-World Instructed Code Edits**
W. Chi*, V. Chen*, R. Shar*, A. Mittal, J.T. Liang, W. Chiang, A.N. Angelopoulos, I. Stoica, G. Neubig, A. Talwalkar, C. Donahue
International Conference on Learning Representations (ICLR) 2026
Acceptance rate: 27%

C13 **Prompts Are Programs Too! Understanding How Developers Build Software Containing Prompts**
J.T. Liang, M. Lin*, N. Rao*, B.A. Myers
ACM International Conference on the Foundations of Software Engineering (FSE) 2025
Acceptance rate: 22%

C12 **Particip-AI: Anticipating Future AI Use Cases and Impacts with Lay Users**
J. Mun, L. Jiang, J.T. Liang, I. Cheong, N. DeCario, Y. Choi, T. Kohno, M. Sap
AAAI/ACM Conference on Artificial Intelligence, Ethics and Society (AIES) 2024
Acceptance rate: 32%

C11 **Can GPT-4 Replicate Empirical Software Engineering Research?**
J.T. Liang, C. Badea, C. Bird, R. DeLine, D. Ford Robinson, N. Forsgren, T. Zimmermann
ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE) 2024
Acceptance rate: 25%

C10 **Counterspeakers' Perspectives: Unveiling Barriers and AI Needs in the Fight against Online Hate**
J. Mun, C. Buerger*, J.T. Liang*, J. Garland, M. Sap
ACM SIGCHI Conference on Human Factors in Computing Systems (CHI) 2024
Acceptance rate: 26%

C9 **A Large-Scale Survey on the Usability of AI Programming Assistants: Successes and Challenges**
J.T. Liang, C. Yang, B.A. Myers
IEEE/ACM International Conference on Software Engineering (ICSE) 2024
Acceptance rate: 25% [Data Skeptic Podcast]

🏆 *Outstanding Paper Award (top 2% of accepted papers)*

C8 **NLPositionality: Characterizing Design Biases of Datasets and Models**
S. Santy*, J.T. Liang*, R. Le Bras, K. Reinecke, M. Sap
Annual Conference of the Association for Computational Linguistics (ACL) 2023
Acceptance rate: 24% [MarkTechPost] [CMU ML Blog]

🏆 *ACM SIGSOFT Distinguished Paper Award (top 5% of accepted papers)*

C7 **A Qualitative Study on the Implementation Design Decisions of Developers**
J.T. Liang, M. Arab, M. Ko, A.J. Ko, T.D. LaToza
IEEE/ACM International Conference on Software Engineering (ICSE) 2023
Acceptance rate: 26% [IEEE Software] [Scholarly Communication Podcast]

C6 **Understanding Skills of OSS Contributors on GitHub**
J.T. Liang, T. Zimmermann, D. Ford
ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE) 2022.
Acceptance rate: 22%

C5 **Towards Mining OSS Skills from GitHub Activity**
J.T. Liang, T. Zimmermann, D. Ford
IEEE/ACM International Conference on Software Engineering—New Ideas and Emerging Results (ICSE-NIER) 2022
Acceptance rate: 27%

C4 **An Exploratory Study of Sharing Strategic Programming Knowledge**
M. Arab, T.D. LaToza, J. Liang, A.J. Ko
ACM SIGCHI Conference on Human Factors in Computing Systems (CHI) 2022
Acceptance rate: 26%

C3 **HowToo: A Platform for Sharing, Finding, and Using Programming Strategies**
M. Arab, J. Liang*, Y. Yoo*, A.J. Ko, T.D. LaToza
IEEE Symposium on Visual Languages and Human–Centric Computing (VL/HCC) 2021
Acceptance rate: 30%

C2 **Whale Watching in Inland Indonesia: Analyzing a Small, Remote, Internet–Based Community Cellular Network**
M. Johnson, J. Liang, M. Lin, S. Singanamalla, K. Heimerl
The Web Conference (WWW) 2021
Acceptance rate: 21%

C1 **Experiences: Design, Implementation, and Deployment of CoLTE, a Community LTE Solution**
S. Sevilla, M. Johnson, P. Kosakanchit, J. Liang, K. Heimerl
ACM International Conference on Mobile Computing and Networking (Mobicom) 2019
Acceptance rate: 19%

Journal Articles

J2 **TableTalk: Scaffolding Spreadsheet Development with a Language Agent**
J.T. Liang, A. Kumar, Y. Bajpai, S. Gulwani, V. Le, C. Parnin, A. Radhakrishna, A. Tiwari, E. Murphy-Hill*, G. Soares*
ACM Transactions of Human-Computer Interaction (TOCHI)
Impact factor: 6.6 [Ploutus]

J1 **An Empirical Investigation of Machines' Capabilities for Moral Judgment with the Delphi Experiment**
L. Jiang, J.D. Hwang, C. Bhagavatula*, R. Le Bras*, J.T. Liang*, S. Levine*, J. Dodge, K. Sakaguchi, M. Forbes, T. Sorenson, J. Borchardt, J. Hessel, S. Gabriel, Y. Tsvetkov, O. Etzioni, M. Sap, R. Rini, Y. Choi
Nature Machine Intelligence, 2024
Impact factor: 18.8 [New York Times] [WIRED] [IEEE Spectrum] [Nature]

Other Refereed Publications (e.g., demos, posters, extended abstracts)

O3 Human-AI Interaction Alignment: Designing, Evaluating, and Evolving Value-Centered AI For Reciprocal Human-AI Futures
H. Shen, T. Knearem, D. Thakkar, P. Pataranutaporn, A. Sinha, J.T. Liang, L. Ahmad, T. Mitra, B.A Myers, Y. Li
ACM SIGCHI Conference on Human Factors in Computing Systems (CHI) 2026—Workshops

O2 LLMs as Workers in Human-Computational Algorithms? Replicating Crowdsourcing Pipelines with LLMs
T. Wu, H. Zhu, M. Albayrak, A. Axon, A. Bertsch, W. Deng, Z. Ding, B. Guo, S. Gururaja, T. Kuo, J.T. Liang, R. Liu, I. Mandal, J. Milbauer, X. Ni, N. Padmanabhan, S. Ramkumar, A. Sudjianto, J. Taylor, Y. Tseng, P. Vaidos, Z. Wu, W. Wu, C. Yang
ACM SIGCHI Conference on Human Factors in Computing Systems (CHI) 2025—Case Studies in Practice

O1 An All–in–One Community LTE Network
S. Sevilla, M. Johnson, P. Kosakanchit, J. Liang, K. Heimerl
ACM International Conference on Mobile Computing and Networking (Mobicom) 2019—Demo Track

Pre-Prints

P4 Learning from 53.6K Real-World Developer Edits of AI-Generated Code
J.T. Liang*, M. Bairathi*, Wayne Chi, Ameet Talwakar, Nishant Subramani, Valerie Chen

P3 AgentBuilder: Exploring Scaffolds for Prototyping User Experiences of Interface Agents
J.T. Liang, T. Barik, J. Nichols, E. Schoop, R. Cheng

P2 Understanding Prompt Programming Tasks and Questions
J.T. Liang, C. Yang, A. Sergeyuk, T. Breaux, B.A. Myers

P1 Navigating Complexity: How Context Shapes Debugging Strategy Choices Among Expert Developers
M. Arab, J.T. Liang, V. Hong, T.D. LaToza

Industry Experience

Apple
Machine Learning Intern, advised by Ruijia Cheng — Summer 2025
Software Engineer Intern — Summer 2019

Microsoft
Applied Scientist Intern, advised by Gustavo Soares — Summer 2024
Research Intern, advised by Thomas Zimmerman & Christian Bird — Summer 2023
Undergraduate Research Intern, advised by Denae Ford Robinson & Thomas Zimmermann — Summer 2021
Software Engineer Intern — Spring 2019
Software Engineer Intern — Summer 2018
Software Engineer Intern — Summer 2017
Intern — Summer 2016

Allen Institute for Artificial Intelligence
Predoctoral Young Investigator, advised by Ronan Le Bras & Yejin Choi — 2021 – 2022
Undergraduate Research Intern, advised by Swabha Swayamdipta & Chandra Bhagavatula — Summer 2020

Uber — Fall 2019
Software Engineer Intern

Research Experience

Carnegie Mellon University 2022 – Now
Graduate Research Assistant, advised by Brad A. Myers (Software & Societal Systems Department)

University of Washington
Undergraduate Research Assistant, advised by Amy J. Ko (Information School) 2020 – 2021
Undergraduate Research Assistant, advised by Spencer Sevilla & Kurtis Heimerl (Computer Science & Engineering) 2019 – 2020

Awards & Honors

Outstanding Paper Award, Annual Conference of the Association for Computational Linguistics 2023. *Awarded to less than 2% of accepted papers / 0.5% of submitted papers.* 2023

ACM SIGSOFT Distinguished Paper Award, IEEE/ACM International Conference on Software Engineering 2023. *Awarded to less than 5% of accepted papers / 1% of submitted papers.* 2023

Ford Fellowship Honorable Mention, Ford Foundation 2023

Graduate Research Fellowship, National Science Foundation ($111,000 over 3 years) 2021

Outstanding Undergraduate Researcher Honorable Mention, Computing Research Association 2021

Undergraduate Service Award, University of Washington, Computer Science & Engineering. *Award to recognize 3 UW CSE undergraduates out of 2,000 students for their outstanding service to the department.* 2020

Husky 100, University of Washington. *Award to recognize 100 UW students out of 55,000 students in honor of their outstanding contributions to the university.* 2020

Outstanding Engineer Award, Allen Institute for Artificial Intelligence ($12,000) 2020

Denice Dee Denton Scholarship ($1,000) 2019

Invited Talks

Prompts Are Programs Too! Understanding How Developers Build Software Containing Prompts
MIT Programming Languages Review Workshop, Cambridge, MA May 2026
East China Normal University, Shanghai, China May 2025
Mila - Quebec Artificial Intelligence Institute, Online Feb 2025

Software Development in the Era of LLMs: Present and Future
JetBrains Research, Online Sep 2024
Microsoft, Online Jul 2024

Selected Press

Scholarly Communication Podcast, "Define Clearly, Select Carefully, End Compellingly" [HTML] Jan 2025

CMU ML Blog, "NLPositionality: Characterizing Design Biases of Datasets and Models" [HTML] Mar 2024

IEEE Software, "Focusing on Developers in the Era of AI and ML" [HTML] Nov 2023

Data Skeptic Podcast, "A Survey Assessing Github Copilot" [HTML] Nov 2023

IEEE Spectrum, "Machines Learn Good From Commonsense Norm Bank" [HTML] Nov 2021

New York Times, "Can a Machine Learn Morality?" [HTML] Nov 2021

Nature, "How robots can learn to follow a moral code" [HTML] Oct 2021

WIRED, "This Program Can Give AI a Sense of Ethics—Sometimes." [HTML] Oct 2021

Teaching

Guest Lectures

Prompt Programming: Past, Present, and Future, Carnegie Mellon University — Mar 2026, Mar 2025
Prompt Engineering, 17-630

Human Aspects of AI-Assisted Software Engineering, North Carolina State University — Oct 2023
Generative AI for Software Engineering, CSC 791-028

Teaching Assistantships

Graduate Teaching Assistant, Carnegie Mellon University, Software & Societal Systems Department — Spring 2024
Foundations of Software Engineering, 17-313. Instructors: Michael Hilton, Eduardo Feo-Flushing

Undergraduate Teaching Assistant, University of Washington, Computer Science & Engineering — Spring 2020
Building Academic Success through Bottom-Up Computing (Introduction to Computer Systems), CSE 390B. Instructors: Dan Grossman, Aaron Johnston, Leslie Ikeda

Undergraduate Teaching Assistant, University of Washington, Computer Science & Engineering — Winter 2020
Software Design & Implementation, CSE 331. Instructor: Hal Perkins

Undergraduate Teaching Assistant, University of Washington, Computer Science & Engineering — Fall 2019
Software Design & Implementation, CSE 331. Instructor: Dan Grossman

Undergraduate Teaching Assistant, University of Washington, Computer Science & Engineering — Winter 2019
Introduction to Programming II, CSE 143. Instructor: Hunter Schafer

Undergraduate Teaching Assistant, University of Washington, Computer Science & Engineering — Fall 2018
Accelerated Introduction to Programming, CSE 143X. Instructor: Stuart Reges

Undergraduate Teaching Assistant, University of Washington, Computer Science & Engineering — Spring 2018
Introduction to Programming II, CSE 143. Instructor: Stuart Reges

Mentoring

Mihika Bairathi, CMU Machine Learning Master's Student — 2025 –
Co-first author on a full paper (P4)

Melissa Lin, CMU Undergraduate Student → Software Engineer @ Google — 2024 –2025
Second author on a full conference paper for FSE 2025 (C13)

Service

Professional Service

Organizer

Organizer, Bi-Align Workshop (2026)
Assistant to the Blending Interaction Subcommittee Chairs, CHI (2026)
ArXiv Track Co-Chair, ACM AIware (2025)
Publicity & Web Chair, ACM AIware (2024)
Student Organizer, PLATEAU (2024, 2025)

PC Member	*Software Engineering:* CHASE (2025, 2026), ICSE – Artifact Track (2025) *Artificial Intelligence*: Bi-Align Workshop (2025), HCI+NLP Workshop (2025)
Reviewer	*Software Engineering:* CACM (2025), EMSE (2024, 2025), TOSEM (2024, 2025), TSE (2025), IEEE Software (2025) *Human-Computer Interaction:* **UIST** (**2024***, **2025***), CHI (2024, 2025, 2026), **CHI—Late Breaking Work** (**2025***) *Artificial Intelligence:* ACL (2025) ***(* = Outstanding Reviewer Award)***
Volunteer	ESEC/FSE (2024), NAACL (2022), Mobicom (2019)

University Service

Student Leader (Ph.D.)	*Student Representative,* CMU S3D Ph.D. Admissions Committee (2025) *Student Representative*, CMU S3D Distinguished Speaker Series Selection Committee (2022, 2023, 2024) *Student Representative*, CMU Software Engineering Ph.D. Curriculum Committee (2022, 2023) *Student Representative*, CMU S3D Software Research Seminar Steering Committee (2022, 2023) *Student Representative*, CMU SCS Dean's Advisory Committee (2022)
Student Leader (Undergrad)	*Member*, UW CSE Student Advisory Council (2021) *Undergraduate Research Leader*, UW Undergraduate Research Program (2021) *Vice Chair*, UW CSE Student Advisory Council (2020) *Officer*, UW CSE Student Advisory Council (2019)